

A Implementation and simulation details

A.1 Response models and assignment implementation

Conditionally on dose and stratum, $y_e = e(d, z) + \varepsilon_e$ and $y_g = g(d, z) + \varepsilon_g$, with independent mean-zero Gaussian errors at the within-combination SDs specified below. The simulations do not generate binary dose-limiting toxicity outcomes. The model-based quantity $\text{PF}(d) = \Pr\{G(d) \leq g^\dagger \mid D_t\} = \Phi\{z_g(d)\}$, where Φ is the standard normal distribution function, concerns latent mean toxicity, not an individual participant’s toxicity probability. The implementation tests $z_g(d) > \Phi^{-1}(\tau)$, which is equivalent to $\text{PF}(d) > \tau$.

Let $\mathcal{E}_t \subseteq \mathcal{G}_t$ be the subset eligible for assignment under the simulated protocol rules. Whenever $A_\tau \cap \mathcal{E}_t$ is nonempty, each acquisition is maximized over that set. Otherwise, the acquisition score does not determine the assignment, and the fallback selects

$$d_t^{\text{fallback}} = \arg \max_{d \in \mathcal{E}_t} z_g(d). \quad (7)$$

Ties are broken by the fixed grid order. Because the fallback combination does not meet the model-based toxicity criterion, Equation (7) is only a simulation-continuation rule; clinical use would require separate safety rules.

GP fitting. Each response has a constant-mean GP with a scaled Matérn-5/2 kernel and a separate lengthscale for each dose axis. After an observed cohort, kernel and mean parameters are fitted by 120 Adam steps on the marginal likelihood with learning rate 0.1. This covariance permits smooth variation over the dose grid but imposes neither monotonicity nor a pharmacologic interaction model. The choice is shared across acquisitions, not selected as clinically optimal. The primary analysis uses the data-generating within-combination outcome variances; these are not estimated from the small simulated trial. $\text{PF}(d)$ uses latent-function variance. Let σ_{ye} and σ_{yg} denote the participant-outcome SDs. For a cohort of two, independent outcome errors contribute $\sigma_{ye}^2/2$ and $\sigma_{yg}^2/2$ to the efficacy and toxicity cohort-mean variances, respectively. These differ from the latent posterior variances $\sigma_e^2(d)$ and $\sigma_g^2(d)$. Hyperparameters remain fixed during each hypothetical update and are re-estimated after the actual simulated cohort.

Independent response models. All rules use separate efficacy and toxicity GPs. We did not study multi-output borrowing, which could change posterior uncertainty and allocation; independence does not imply equivalence to a joint model^{32;33}.

A.2 What the allocation scores measure

The four allocation scores value different information about dose response. For cKG, we can separate the effects of efficacy learning and changes in which combinations meet the toxicity criterion. For a candidate next-cohort combination q , let C_t be the current value set defined in Section 2.2, and let C_t^+ be the corresponding set after a hypothetical cohort. Adding and subtracting the updated efficacy maximum over C_t gives

$$\alpha_{\text{cKG}}(q) = \left[\mathbb{E} \left\{ \max_{d \in C_t} \mu_e^+(d) \right\} - \max_{d \in C_t} \mu_e(d) \right] \quad (8)$$

$$+ \mathbb{E} \left\{ \max_{d \in C_t^+} \mu_e^+(d) - \max_{d \in C_t} \mu_e^+(d) \right\}. \quad (9)$$

The first term is nonnegative by the posterior martingale property and Jensen’s inequality, conditional on the fitted GP and fixed hyperparameters. It measures efficacy learning with the current selection set fixed. The second term accounts for changes to that set after the toxicity update and can have either sign. A negative cKG score can therefore reflect a decrease in the value of this constrained selection rule. It need not indicate numerical error or imply that information has negative value in an unrestricted decision problem.

cEI weights local expected efficacy improvement by $\text{PF}(q)$. tMSE depends only on $z_g(q)$ and $\sigma_g(q)$, so candidates with the same local toxicity summaries receive the same score even if they inform other combinations differently. The entropy score sums the mutual information between each indicator $\mathbf{1}\{G(x) \leq g^\dagger\}$ and the new toxicity observation at q , conditional on D_t . It values classification information equally at each currently available grid combination and has no efficacy term. cKG instead carries both response updates through the final-selection rule.

Increasing τ from τ_1 to τ_2 removes, for a fixed posterior, combinations with $\Phi^{-1}(\tau_1) < z_g(d) \leq \Phi^{-1}(\tau_2)$. For cKG, this also changes which hypothetical toxicity updates alter the selection set. For cEI, the local weight remains $\Phi\{z_g(q)\}$ among eligible candidates. Neither formula implies a fixed ordering of cKG and cEI across cutoffs. Changing τ also changes the data collected over a trial; nesting the selection sets for a fixed posterior does not guarantee an ordering of realized allocation or selection errors.

A.2.1 Deterministic evaluation of cKG on a finite dose set

Fix a query q and index the combinations in $\mathcal{G}_t$ by i , with dose d_i . Write $\mu_{e,i} = \mu_e(d_i)$ and $\mu_{g,i} = \mu_g(d_i)$. Conditional on the current GP fit, the updated posterior means have the affine form below, using independent $Z_e, Z_g \sim \mathcal{N}(0, 1)$:

$$\mu_{e,i}^+ = \mu_{e,i} + b_{e,i}Z_e, \quad \mu_{g,i}^+ = \mu_{g,i} + b_{g,i}Z_g, \quad (10)$$

Here $b_{e,i} = \text{Cov}\{E(d_i), \bar{Y}_e(q) \mid D_t\} / \sqrt{\text{Var}\{\bar{Y}_e(q) \mid D_t\}}$, with $b_{g,i}$ defined analogously for toxicity. For numerical stability, the implementation uses the floored post-update latent toxicity variance

$$(s_{g,i}^+)^2 = \max\left[\sigma_g^2(d_i) - \frac{\text{Cov}\{G(d_i), \bar{Y}_g(q) \mid D_t\}^2}{\text{Var}\{\bar{Y}_g(q) \mid D_t\}}, 10^{-12}\right].$$

Thus every hypothetical standardized toxicity margin is an affine line in the same scalar variable,

$$z_{g,i}^+(Z_g) = a_i + c_i Z_g, \quad a_i = \frac{g^\dagger - \mu_{g,i}}{s_{g,i}^+}, \quad c_i = -\frac{b_{g,i}}{s_{g,i}^+}. \quad (11)$$

Write $q_\tau = \Phi^{-1}(\tau)$, and let φ denote the standard normal density. The finite evaluator follows four steps.

1. The crossings of $a_i + c_i Z_g = q_\tau$ partition the real line into intervals on which the set $S = \{i : z_{g,i}^+(Z_g) > q_\tau\}$ is fixed.
2. On an interval with $S \neq \emptyset$, the conditional value is $\mathbb{E}_{Z_e}[\max_{i \in S} \{\mu_{e,i} + b_{e,i}Z_e\}]$. Sorting slopes and removing dominated lines gives its upper-envelope segments. If line i is active on (ℓ, u) , its contribution is

$$\mu_{e,i}\{\Phi(u) - \Phi(\ell)\} + b_{e,i}\{\varphi(\ell) - \varphi(u)\}. \quad (12)$$

3. On an interval with $S = \emptyset$, pairwise crossings of the lines in Equation (11) partition the greatest- z_g fallback. If candidate j is selected on a segment, independence gives $\mathbb{E}_{Z_e}(\mu_{e,j} + b_{e,j}Z_e) = \mu_{e,j}$.

4. Each conditional value is multiplied by its standard-normal Z_g interval probability and the pieces are summed. Subtracting V_t gives $\alpha_{\text{cKG}}(q)$; centering the efficacy intercepts at V_t before integration avoids subtracting large offset-shifted values.

The final-selection value is evaluated over all currently available combinations in $\mathcal{G}_t$; $\mathcal{E}_t$ restricts the next assignment but not the combinations entering that value. Conditional on the 10^{-12} variance floor, finite $\mathcal{G}_t$, independent Gaussian response channels, and fixed hyperparameters during the hypothetical update, the interval decomposition computes the one-cohort score deterministically, subject only to floating-point arithmetic. It does not project later cohorts, future dose availability, or stopping.

A.3 Numerical validation

We checked the deterministic cKG calculation on 720 posterior states comprising 18,000 query evaluations. On a stratified 64-state subset, eight independently scrambled Sobol approximations per state (512 checks in total), each using 131,072 points, differed from the deterministic scores by at most 9.40×10^{-5} .

For the entropy rule, we used nine-point Gauss–Hermite quadrature, which approximates a Gaussian expectation by a weighted sum at selected integration points. A finer 128-point calculation gave the same dose choices in the checked cases where at least one combination met the toxicity criterion. This comparison does not establish a general integration error bound.

We independently checked the sensitivity archive’s observations and analyses. All 24,000 paths and their 72,000 nested snapshots were recovered. Recorded random seeds reproduced the Gaussian outcome innovations and initialization locations. Dense Matérn-5/2 conditioning with each snapshot’s saved fitted parameters reproduced its posterior predictions; the largest absolute discrepancy in a posterior mean was below 4×10^{-9} . Recomputed means, MCSEs, conditional denominators, and interval endpoints agreed with all 118 primary and 1,227 secondary summaries. These are checks of recorded paths and fixed-parameter calculations, not new adaptive trials or independent GP hyperparameter refits.

We compared maintained software candidate 0.2.1rc4 with the frozen study implementation in one numerical environment, using 31 prospectively specified matched configurations. Shared posterior, score, allocation, and outcome fields agreed within the prespecified numerical tolerances in 30 pairs, including all 24 full-budget configurations drawn from the paper settings. The candidate corrects cancellation when computing normal probabilities over narrow intervals. Relative to the preceding candidate (0.2.1rc3), all 31 configurations retained identical fitted parameters, allocation paths, and terminal results; the largest absolute cKG score change was 4.44×10^{-16} . In one short-budget gradual-availability tMSE pair, final selection changed because the maintained implementation restricts that choice to combinations already opened; the fitted models and allocation path were identical. This boundary correction does not change the reported full-budget design, whose grid is fully opened before completion; stopped trials make no final selection. This comparison checks software maintenance; it does not re-estimate operating characteristics.

The maintained candidate also validates inputs and controller state, reports final-selection status explicitly, and versions checkpoint keys. The reported 40-record gradual-availability simulations open the entire grid before scheduled completion, so the restriction above does not affect their results. Reported operating characteristics refer to the archived study source. Clinical use requires a separately specified and validated complete trial design.

A.4 Fixed-data computation times

We benchmarked the installed pre-rename software candidate 0.2.1rc4 on three prespecified, nested OSA stratum-0 datasets with 4, 20, and 40 on-grid observations, generated nonadaptively with fixed seed 20261001. Each dataset used the same two fitted GPs for all four rules, with 25 available combinations, $\tau = 0.7$, and cohort size two. Table 5 reports the median of five scoring repetitions after one warm-up. Scoring includes construction of the gate and shared context, with a fresh context and no score reuse; fitted-GP prediction caches were warm. GP fitting was measured once per dataset after a separate warm-up pair, with 120 Adam steps per response. Execution was serial on an Apple M4 CPU with 16 GB memory and numerical-library and PyTorch thread limits set to one. The benchmark script, fixed observations, scores and individual timings are retained alongside the maintained software source. These timings measure fixed-data operations and exclude adaptive trial simulation. The current package is named `dose-combination-bo`; renaming it does not change the numerical implementation.

Table 5: Computation times for the installed software on fixed OSA datasets.

Records	Gate-admissible combinations	Two-GP fitting (s)	Median scoring time (ms)			
			cKG	tMSE	Entropy	cEI
4	25	0.136	112.0	0.423	12.90	14.95
20	10	0.136	104.7	0.446	13.03	14.87
40	12	0.147	113.9	0.421	13.21	15.08

Note. Each rule returns a score vector on the 25-combination grid. Fitting times are single measurements for the pair of response models, rather than medians or full-trial runtimes.

A.5 Simulation settings and source context

Obstructive sleep apnea (OSA)-derived simulation specification. The OSA-derived setting uses standardized dose $d = (d_1, d_2) \in [0, 1]^2$, stratum $z \in \{0, 1\}$, and the seven-term column basis $b(d) = (1, d_1, d_2, d_1 d_2, d_1^2, d_2^2, d_1^2 d_2^2)^\top$. The implementation follows the polynomial specification in archived version 1 of Willard et al.¹², with $f = -e$, latent toxicity score g , and the rounded coefficient vectors reported there:

$$\begin{aligned}
 f(d, z) &= \beta_z^\top b(d), & g(d, z) &= \theta^\top b(d), \\
 \beta_0 &= (-1.38, -4.08, -0.48, -4.23, 2.45, -7.51, -1.56)^\top, \\
 \beta_1 &= (1.05, -11.28, -8.32, -17.02, 8.17, 2.34, 4.61)^\top, \\
 \theta &= (-0.59, 1.83, 2.26, -4.05, 1.79, 0.47, 2.91)^\top \quad (\text{shared}).
 \end{aligned}$$

The toxicity thresholds are $g^\dagger(0) = 1.5$ and $g^\dagger(1) = 2.0$; fixed within-combination SDs are $\sigma_{ye} = 7.68$ and $\sigma_{yg} = 1.29$ in both strata. On a fixed 200×200 grid, the continuous constrained optima under these rounded coefficients are approximated by $(0.337, 0.819)$ and $(0.543, 0.874)$ and lie numerically on the toxicity boundary. The thresholds and outcome SDs are simulator inputs, not clinically calibrated cutoffs or empirical MARIPOSA estimates.

Source context. Willard et al.¹² constructed the OSA-derived simulator from published arm-level changes in the 4%-desaturation apnea-hypopnea index (AHI4) and adverse-event frequencies, using unvalidated weights to form a synthetic burden score. The strata represent mild/moderate ($z = 0$) and severe ($z = 1$) OSA. Toxicity inherits the natural-log scale $\log(B + 0.5)$, where B is the weighted

adverse-event burden. The present simulator generates Gaussian responses with polynomial mean $g(d, z)$ on this scale; it does not generate individual adverse-event burdens. No patient-level records were fitted here. In MARIPOSA, aroxybutynin varied only with atomoxetine fixed at 75 mg, so the simulator’s remaining two-dimensional surface is model-generated extrapolation rather than a reconstruction of the trial¹⁸.

Logistic and MARIPOSA-motivated simulation specifications. The disease-agnostic logistic scenario uses $L(x) = \{1 + \exp(-x)\}^{-1}$,

$$e(d, z) = L(-1 + 2.6d_1 + 2.2d_2 - 1.1d_1^2 - 0.9d_2^2 + 0.3d_1d_2),$$

and $g(d, z) = L\{a_{0z} + a_{1z}d_1 + a_{2z}d_2 + a_{12,z}d_1d_2\}$, with coefficients $(-2, 3, 2.5, 1)$ for $z = 0$ and $(-2, 2, 1.8, 0.5)$ for $z = 1$, and $g^\dagger = 0.30$. Both outcomes increase monotonically with either dose, efficacy is bounded on the logistic scale, and the narrow feasible region places the constrained optimum near the toxicity threshold. On an endpoint-inclusive 80×80 grid, the continuous-domain constrained-optimum approximations are $(0.190, 0.215)$ and $(0.304, 0.278)$; the within-combination efficacy SD is 0.24825, and the toxicity-outcome SDs are 0.32188 and 0.27135. These probability-shaped surfaces are observed with Gaussian error; they are neither binary endpoints nor a named disease model.

The synthetic MARIPOSA-motivated surface, identical in both strata, is

$$e(d) = 50[1 - \exp\{-(1.5d_1 + 0.6d_2 + 1.8d_1d_2)\}], \quad g(d) = 0.5 + 1.4d_1 + 0.5d_2 + 0.6d_1d_2, \quad g^\dagger = 2.0.$$

For this surface, the continuous constrained optimum is $(0.5, 1.0)$, which is also on the 5×5 assignment grid. To see this, maximize the increasing efficacy index $1.5d_1 + 0.6d_2 + 1.8d_1d_2$ subject to $1.4d_1 + 0.5d_2 + 0.6d_1d_2 \leq 1.5$. For $d_1 \leq 0.5$ the best feasible d_2 is 1; for $d_1 \geq 0.5$ it is $(1.5 - 1.4d_1)/(0.5 + 0.6d_1)$, along which the index decreases. The within-combination efficacy and toxicity-outcome SDs are 19.6720 and 0.76931. They were chosen to give signal-to-noise ratios similar to OSA, not estimated from clinical data. The author-specified coefficients create saturating efficacy and increasing burden; they do not estimate a two-dimensional clinical surface or establish pharmacologic synergy.

A.6 Finite-grid targets

Table 6 evaluates Equation (1) directly on the assignment grid. These deterministic values use the rounded functions and thresholds above and in Appendix C; they are not estimates from simulated trials. A fine evaluation grid may omit assignment doses and therefore does not define the assignment-grid target.

A.7 Outcome definitions

A combination is above the toxicity limit when its true mean satisfies $g(d, z) > g^\dagger(z)$. For above-limit final selection, the all-trial percentage divides the number of above-limit selections by all initiated trials; stopped trials contribute no above-limit selection. The selection-conditional percentage and mean efficacy use only trials that make a selection. These conditional summaries are undefined if no trial selects, and no efficacy value is assigned to stopped trials. For the full-grid design, let d_i be the combination received by post-initialization participant i and define

$$N_{\text{above}}^{\text{post}} = \sum_{i=1}^{36} \mathbf{1}\{g(d_i, z) > g^\dagger(z)\}.$$

Table 6: True constrained targets on the 5×5 assignment grid.

Scenario	Stratum	Acceptable			
		pairs / 25	d_{grid}^*	$e(d_{\text{grid}}^*)$	$g(d_{\text{grid}}^*)$
OSA-derived	0	11	(0.25, 0.75)	7.6792	1.2817
OSA-derived	1	16	(0.50, 0.75)	13.2055	1.6223
Logistic	0	3	(0.25, 0.00)	0.3968	0.2227
Logistic	1	6	(0.25, 0.25)	0.5234	0.2653
MARIPOSA-motivated	Both	18	(0.50, 1.00)	44.7300	2.0000
Gaussian bump	0	10	(0.25, 0.50)	2.6855	0.3027
Gaussian bump	1	10	(0.50, 0.50)	3.9088	1.7000

We report $100N_{\text{above}}^{\text{post}}/36$, the percentage assigned above the limit after initialization, as a main allocation outcome. For gradual availability, N_{above} counts above-limit assignments among all enrolled participants, including the initial cohort. We report its mean per initiated trial together with enrollment and stopping. Both counts concern assignments to combinations with above-limit true mean toxicity, not observed adverse events.

Alongside efficacy and above-limit selection, Appendix A.11 reports the probability of a truly acceptable selection within a specified efficacy margin of d_{grid}^* in OSA. A zero margin gives exact acceptable grid-optimal selection.

A.8 Additional operating characteristics

The main-text Table 1 reports absolute OSA-derived results. Figure 3 below shows the surface-by- τ contrasts.

Table 7: cKG-minus-comparator differences in true mean efficacy at final selection under full-grid availability.

Scenario	Comparator	Efficacy difference (MCSE) [95% interval]
OSA-derived	tMSE	0.104 (0.109) [−0.110, 0.319]
OSA-derived	Entropy rule	−0.563 (0.119) [−0.797, −0.329]
Logistic	tMSE	0.024 (0.004) [0.017, 0.031]
Logistic	Entropy rule	0.024 (0.004) [0.017, 0.031]
MARIPOSA-motivated	tMSE	−0.508 (0.336) [−1.176, 0.159]
MARIPOSA-motivated	Entropy rule	0.115 (0.463) [−0.804, 1.034]
Gaussian bump	tMSE	0.165 (0.035) [0.096, 0.235]
Gaussian bump	Entropy rule	−0.074 (0.041) [−0.156, 0.007]

Note. Differences average two strata and five τ settings equally within each paired replicate. Intervals are pointwise, unadjusted 95% Monte Carlo t intervals. OSA-derived and Gaussian-bump scenarios use $M = 200$; the others use $M = 100$. Positive values favor cKG. All trials made a final selection, including fallback selections. Efficacy scales differ across scenarios.

A.9 Final-selection efficacy relative to cEI

Table 8 gives the complete τ profile for cKG versus cEI.

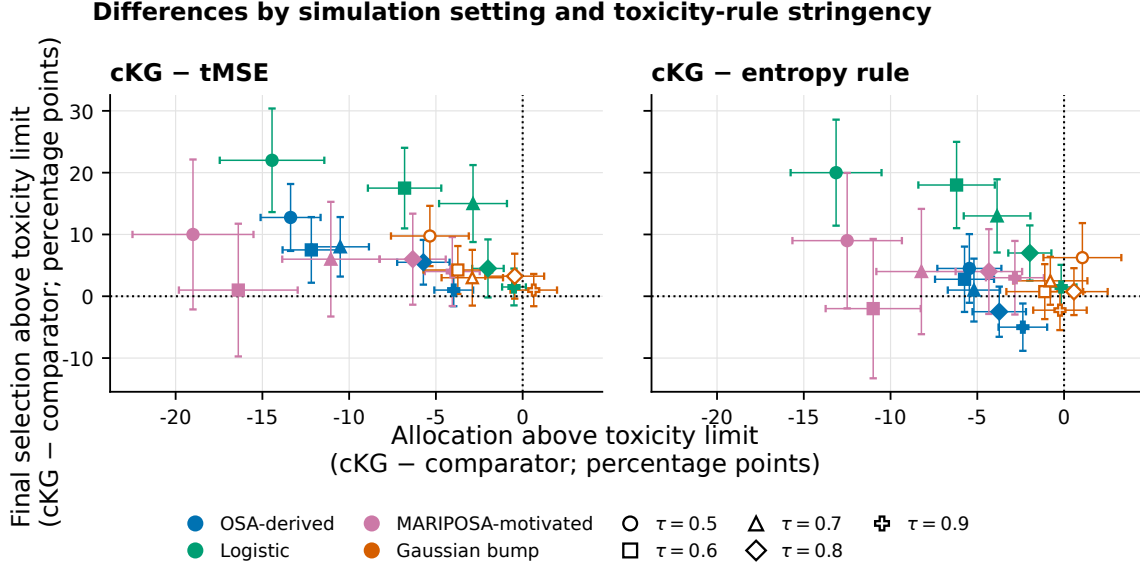


Figure 3: cKG-minus-comparator differences in above-limit allocation (horizontal) and final selection (vertical) under full-grid availability. The left and right panels compare cKG with tMSE and the entropy rule, respectively. Each point is one surface- τ setting, averaged over two strata; bars are pointwise unadjusted 95% Monte Carlo intervals. The upper-left quadrant denotes fewer above-limit assignments but more above-limit final selections under cKG. OSA-derived and Gaussian-bump settings use $M = 200$ replicates; the others use $M = 100$.

Table 8: cKG-minus-cEI differences in true mean efficacy at final selection, by τ , under full-grid availability.

τ	OSA-derived	Logistic	MARIPOSA-motivated	Gaussian bump
0.5	1.192 [0.755, 1.630]	0.0374 [0.0208, 0.0539]	6.160 [5.102, 7.217]	0.389 [0.240, 0.538]
0.6	1.034 [0.622, 1.447]	0.0338 [0.0190, 0.0486]	6.188 [5.041, 7.335]	0.283 [0.143, 0.423]
0.7	0.905 [0.505, 1.305]	0.0245 [0.0109, 0.0381]	4.172 [2.756, 5.589]	0.174 [0.054, 0.295]
0.8	0.391 [0.040, 0.743]	0.0091 [-0.0024, 0.0206]	3.463 [2.007, 4.919]	0.044 [-0.043, 0.130]
0.9	0.172 [-0.125, 0.468]	0.0080 [-0.0013, 0.0173]	3.092 [1.534, 4.649]	0.063 [-0.022, 0.147]

Note. Entries are mean differences [pointwise unadjusted 95% Monte Carlo t interval], weighting the two strata equally; positive values indicate higher true mean efficacy under cKG. OSA-derived and Gaussian-bump analyses use $M = 200$ replicates, and the others use $M = 100$. Efficacy scales differ by setting. Results include prespecified fallback selections; in the Gaussian-bump analysis, these occurred in 17.25–41.75% of cKG trials and 17.25–41.25% of cEI trials across τ .

A.10 Whether a final selection passed the toxicity criterion

Table 9 partitions the original final selections without changing or rerunning either policy. These exploratory summaries distinguish four mutually exclusive categories: criterion-passing and truly acceptable, criterion-passing and truly above limit, fallback and truly acceptable, or fallback and truly above limit. The categories use all initiated full-grid trials as the denominator. Their sum is 100% within each row because these runs always made a final selection. Adding a no-selection option would require specifying its value, revising the lookahead, and running new adaptive simulations. It cannot be evaluated by relabeling these recorded selections.

Table 9: Final-selection status and true mean toxicity for cKG and cEI (percent of all trials).

Scenario	Rule	Passed posterior criterion		Fallback selection	
		Acceptable	Above limit	Acceptable	Above limit
OSA-derived	cKG	80.55	18.55	0.65	0.25
	cEI	88.30	10.15	1.20	0.35
Logistic	cKG	71.70	21.80	5.20	1.30
	cEI	81.70	10.70	5.70	1.90
MARIPOSA-motivated	cKG	79.60	20.40	0.00	0.00
	cEI	99.80	0.20	0.00	0.00
Gaussian bump	cKG	63.70	9.40	16.75	10.15
	cEI	62.95	10.55	16.80	9.70

Note. Each row averages two strata and five τ values equally. Acceptable means $g(\hat{d}, z) \leq g^\dagger(z)$ in the simulation truth, not a demonstrated clinical safety claim. Counts are 2,000 trials per policy in OSA and Gaussian bump and 1,000 in each other scenario. These are descriptive post hoc partitions of the same records used for the main contrasts; they are not new simulation experiments.

Machine-readable files include conditional efficacy and the corresponding denominators. Because these subsets differ across policies, their summaries do not measure unconditional policy effects. In Gaussian bump, for example, criterion-passing selections have mean efficacy 1.433 under cKG (1,462 trials) and 1.189 under cEI (1,470 trials), whereas fallback selections have means 0.486 (538 trials) and 0.430 (530 trials). Reporting only the criterion-passing means would omit more than one quarter of the original trials.

A.11 Acceptable selection within an efficacy margin

Using the recorded full-grid OSA selections, we estimated the probability of selecting a combination with true mean toxicity at or below its limit and true mean efficacy within δ of $e(d_{\text{grid}}^*, z)$. The reference is the best truly acceptable combination on the 25-point assignment grid, not a continuous-domain optimum. Before deriving these post hoc summaries, we fixed $\delta \in \{0, 0.5, 1, 2\}$ simulated AHI4 events/hour as a descriptive sensitivity range, without clinical calibration. All initiated trials contribute, retaining fallback selections and evaluating them by the same truth-based criterion.

The 0.5- and 1-event margins give the same rates as zero because neither stratum has another acceptable grid combination within one event/hour of its optimum. cKG had higher estimated rates than cEI at every cutoff and margin. Entropy had higher estimated rates than cKG throughout; the cKG–tMSE ranking depended on the cutoff and margin. These summaries complement mean efficacy and above-limit selection; they do not establish a clinical benefit–risk ranking.

Table 10: Truly acceptable selections within an efficacy margin of the grid optimum in OSA, under full-grid availability.

τ	Rule	$\delta = 0$	$\delta = 0.5$	$\delta = 1$	$\delta = 2$
0.5	cKG	21.00 (2.08)	21.00 (2.08)	21.00 (2.08)	38.00 (2.41)
0.5	tMSE	24.75 (2.30)	24.75 (2.30)	24.75 (2.30)	46.25 (2.58)
0.5	Entropy rule	22.25 (2.09)	22.25 (2.09)	22.25 (2.09)	41.75 (2.62)
0.5	cEI	17.00 (1.99)	17.00 (1.99)	17.00 (1.99)	36.00 (2.63)
0.6	cKG	18.25 (2.04)	18.25 (2.04)	18.25 (2.04)	34.50 (2.41)
0.6	tMSE	19.50 (1.97)	19.50 (1.97)	19.50 (1.97)	36.75 (2.42)
0.6	Entropy rule	21.00 (2.05)	21.00 (2.05)	21.00 (2.05)	38.50 (2.50)
0.6	cEI	10.25 (1.52)	10.25 (1.52)	10.25 (1.52)	30.25 (2.60)
0.7	cKG	14.25 (1.71)	14.25 (1.71)	14.25 (1.71)	28.75 (2.23)
0.7	tMSE	13.75 (1.81)	13.75 (1.81)	13.75 (1.81)	30.50 (2.50)
0.7	Entropy rule	14.50 (1.86)	14.50 (1.86)	14.50 (1.86)	32.00 (2.48)
0.7	cEI	7.25 (1.39)	7.25 (1.39)	7.25 (1.39)	21.00 (2.08)
0.8	cKG	9.75 (1.61)	9.75 (1.61)	9.75 (1.61)	23.25 (2.06)
0.8	tMSE	11.00 (1.59)	11.00 (1.59)	11.00 (1.59)	18.00 (2.13)
0.8	Entropy rule	11.00 (1.59)	11.00 (1.59)	11.00 (1.59)	24.25 (2.18)
0.8	cEI	7.00 (1.28)	7.00 (1.28)	7.00 (1.28)	17.25 (1.99)
0.9	cKG	5.75 (1.29)	5.75 (1.29)	5.75 (1.29)	13.50 (1.83)
0.9	tMSE	5.50 (1.22)	5.50 (1.22)	5.50 (1.22)	10.75 (1.66)
0.9	Entropy rule	9.75 (1.53)	9.75 (1.53)	9.75 (1.53)	17.25 (1.99)
0.9	cEI	5.50 (1.22)	5.50 (1.22)	5.50 (1.22)	12.00 (1.63)

Note. Entries are percentages (MCSE in pp), averaging the two stratum indicators within each of $M = 200$ independent replicate sets. Each rule–cutoff row uses all 400 initiated trials, retaining fallback selections. Acceptability uses true mean toxicity, not the posterior criterion. Margins are in simulated AHI4 events/hour and are descriptive, not clinically calibrated. $\delta = 0$ denotes selection of the exact acceptable grid optimum. These post hoc summaries reuse existing records; no adaptive trials were rerun.

A.12 Dose availability and stopping sensitivity

The OSA-derived gradual-availability analysis is motivated by Willard et al.¹² but is not a reconstruction. The two strata are analyzed separately; no shared accrual, cross-stratum borrowing, or enrollment reallocation is modeled.

The gradual design begins with two participants at (0,0). After cohort t , the grid available for the next cohort is

$$\mathcal{G}_t = \{d \in \mathcal{G} : d_1 + d_2 \leq 0.25(t + 1)\}.$$

The full grid is available before the eighth cohort. Before then, previously assigned combinations are ineligible while an available unassigned combination remains. On the first two consecutive occasions when no eligible combination meets $\text{PF}(d) > \tau$, Equation (7) supplies the fallback; a third such occasion stops the simulated trial before assignment and without a final selection. The count resets when an eligible combination meets the requirement. At maximum enrollment, the final rule uses the full grid. Dose availability follows this fixed schedule rather than observed outcomes. Permanent elimination, delayed outcomes, independent monitoring, and acquisition-value stopping are not modeled.

The analysis includes both strata, the four reported rules, $M = 200$ independent replicates, and $\tau = 0.7, 0.9$. Within a replicate, rules share random-number streams for initialization and outcome errors; each rule– τ setting therefore contains 400 initiated stratum-specific trials. Selection-conditional contrasts include only matched pairs in which both rules make a final selection; other contrasts include all matched pairs. Intervals are pointwise 95% Monte Carlo t intervals. Main-paper Table 4 reports the operating characteristics and denominators.

B Sensitivity to participant count and assumed toxicity-outcome variability

B.1 Scope, paths, and snapshot estimands

The nested sensitivity analysis uses the OSA-derived surface in two strata, the full 5×5 dose set, $\tau \in \{0.7, 0.9\}$, four initialization observations, cohorts of two, and no stopping. It compares cKG with tMSE, the entropy rule, and cEI.

Efficacy and toxicity outcomes are generated with fixed within-combination SDs of 7.68 and 1.29, respectively. The analysis model uses 7.68 for efficacy and 0.645, 1.29, or 2.58 for toxicity. These SDs remain fixed throughout each simulated trial. Thus only the assumed toxicity-outcome SD changes, through

$$c_\sigma = \frac{\text{toxicity-outcome SD specified in the analysis}}{\text{toxicity-outcome SD used to generate the data}} \in \{0.5, 1, 2\}.$$

The three values understate, match, or overstate within-combination toxicity-outcome variability. The data-generating surfaces and outcome SDs, efficacy model, and all other analysis choices remain fixed. Settings share Gaussian innovations within replicate, although adaptive assignments can produce different dose–outcome histories. This analysis therefore isolates one likelihood assumption and does not address binary DLTs or other forms of model misspecification.

Each simulated trial continues to 80 response records and is analyzed after $R \in \{20, 40, 80\}$ records. Because four records are initialization observations, the analyses follow $R - 4 \in \{16, 36, 76\}$ participant assignments. At snapshot R , let $N_{\text{above}}^{\text{post}}$ count above-limit assignments among those $R - 4$ participants, and let $\hat{d}_R$ be the combination selected using the first R records. Allocation is summarized as $100N_{\text{above}}^{\text{post}}/(R - 4)$. Above-limit final selection is reported as the percentage of trials with $g(\hat{d}_R) > g^\dagger$. The response models use the initialization records, but allocation summaries exclude them. These participant counts are not clinical sample-size recommendations; M denotes independent simulation replicates, not enrollment.

B.2 Contrast families and uncertainty rules

Each outcome is averaged equally over the two strata and two τ settings within replicate. The displayed family contains 54 cKG-minus-comparator contrasts: three comparators, two outcomes, three snapshots, and three c_σ settings. The replicate is the independent Monte Carlo unit. Simultaneous 95% max- t intervals use the higher empirical 0.95 quantile of the maximum absolute studentized coordinate from 100,000 Rademacher multiplier draws.

Replication began at $M = 200$ and all settings were extended together to $M = 500$ under a prespecified precision screen^{29–31}. The cap of 500 was set before the $M = 500$ effects were examined. At that cap, 51 of the 54 displayed contrasts met the 1.5-pp MCSE criterion. The three exceptions were above-limit final-selection contrasts with overstated toxicity variability: cKG versus the entropy rule after 20 and 40 records and cKG versus cEI after 20 records; the largest MCSE was 1.744 pp. The complete 118-row precision screen is retained in the reproducibility files. The analysis does not establish a clinical sample size or robustness to other forms of model misspecification.

B.3 Acquisition comparisons across snapshots and toxicity-outcome variability

Figure 4 displays all 54 cKG-minus-comparator contrasts summarized in the main paper.

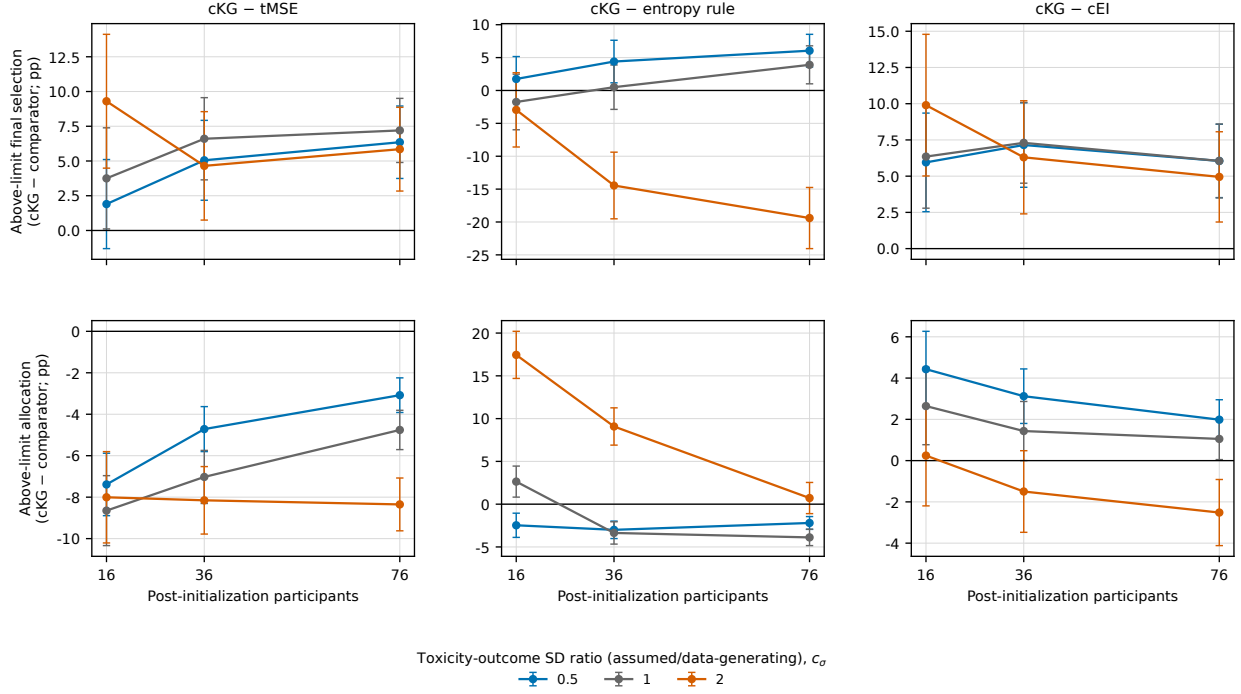


Figure 4: cKG-minus-comparator differences across participant counts and assumed toxicity-outcome variability in the OSA-derived analysis ($M = 500$). Columns identify the comparator; rows show above-limit final selection and post-initialization allocation. Colors show the ratio of the assumed to the data-generating toxicity-outcome SD. Points weight the two strata and $\tau = 0.7, 0.9$ equally; this average is descriptive. The 16, 36, and 76 assignments correspond to 20, 40, and 80 response records. Bars are simultaneous 95% max- t Monte Carlo intervals across the 54 contrasts. Positive values indicate more above-limit final selections or assignments under cKG.

Table 11 reports final-selection efficacy for the same nine SD-by-snapshot settings. These are descriptive means over all trials, including fallback selections; there was no stopping. These summaries are separate from the family of 54 toxicity contrasts.

Table 11: Mean true efficacy at final selection across assumed toxicity SDs and record counts.

c_σ	Records R	cKG	tMSE	Entropy	cEI
0.5	20	6.43 (0.12)	6.60 (0.12)	6.89 (0.11)	6.12 (0.11)
0.5	40	6.52 (0.11)	6.45 (0.10)	6.68 (0.10)	5.94 (0.10)
0.5	80	6.68 (0.10)	6.46 (0.09)	6.75 (0.09)	5.98 (0.09)
1	20	6.56 (0.12)	6.34 (0.11)	7.19 (0.12)	5.98 (0.11)
1	40	6.73 (0.10)	6.49 (0.10)	7.24 (0.11)	6.15 (0.10)
1	80	7.08 (0.10)	6.79 (0.09)	7.34 (0.10)	6.24 (0.10)
2	20	6.98 (0.15)	6.01 (0.14)	8.68 (0.15)	6.38 (0.14)
2	40	6.81 (0.12)	6.44 (0.12)	8.90 (0.15)	6.49 (0.12)
2	80	6.92 (0.10)	6.92 (0.09)	9.09 (0.13)	6.70 (0.10)

Note. Entries are means (MCSEs), averaging two strata and $\tau = 0.7, 0.9$ equally within each of $M = 500$ replicates. Four initialization records are included in R . The ratio c_σ compares the assumed toxicity-outcome SD with the SD used to generate the data. Efficacy is simulated AHI4 reduction in events/hour. These marginal summaries do not provide paired uncertainty for differences between rules.

C Gaussian-bump simulation specification

The Gaussian-bump scenario has nonmonotone mean toxicity and Gaussian-shaped peaks in efficacy. Its continuous-domain unconstrained efficacy maximum has mean toxicity below the limit in both strata; on the 5×5 assignment grid this holds only in stratum 1. Each stratum has 10 acceptable combinations on the assignment grid. In stratum 0, mean toxicity increases from 0.303 at $(0.25, 0.50)$ to 1.250 at the adjacent pair $(0.50, 0.50)$. The lowest pair $(0, 0)$ exceeds the limit in both strata; these full-grid runs do not use it as a required initial assignment. Figure 5 shows the geometry. The same full-grid algorithms and outcome measures are used, with $M = 200$ replicates and one simulated trial per stratum in each replicate.

The exact data-generating functions are

$$\begin{aligned}
 e(d, 0) &= 3.2 \exp\{ -[(d_1 - 0.3)^2/0.03 + (d_2 - 0.7)^2/0.1] \} \\
 &\quad + \sin(2\pi d_1) \cos(3\pi d_2) + 1.4d_1 \exp(-1.2d_1^2) + 0.9d_2 \exp(-0.6d_2^2), \\
 e(d, 1) &= 2 \exp\{ -[(d_1 - 0.5)^2 + (d_2 - 0.5)^2]/0.05 \} \\
 &\quad + 1.2 \sin(3\pi d_1) \sin(3\pi d_2) + d_1 \exp(-d_1^2) + 0.8d_2 \exp(-0.9d_2^2), \\
 g(d, 0) &= 3[(d_1 - 0.5)^2 + (d_2 - 0.5)^2] + 0.5 \sin(3\pi d_1) \sin(3\pi d_2) + 1.5(d_1^2 + d_2^2), \\
 g(d, 1) &= 4[(d_1 - 0.5)^2 + (d_2 - 0.5)^2] + 0.7 \sin(3\pi d_1) \sin(3\pi d_2) + 2(d_1^2 + d_2^2).
 \end{aligned}$$

On the endpoint-inclusive 80×80 evaluation grid, the median toxicity defines $g^\dagger = 1.29633$ and 1.73722 in strata 0 and 1. The corresponding continuous-domain constrained-optimum approximations are $(0.291, 0.684)$ and $(0.506, 0.506)$. Efficacy-outcome SDs are 1.86599 and 1.48427; toxicity-outcome SDs are 1.19728 and 1.60591. These fixed values are not empirical disease-specific variability estimates. Because several surface features differ from OSA, these comparisons cannot isolate the effect of multimodality or any other single feature.

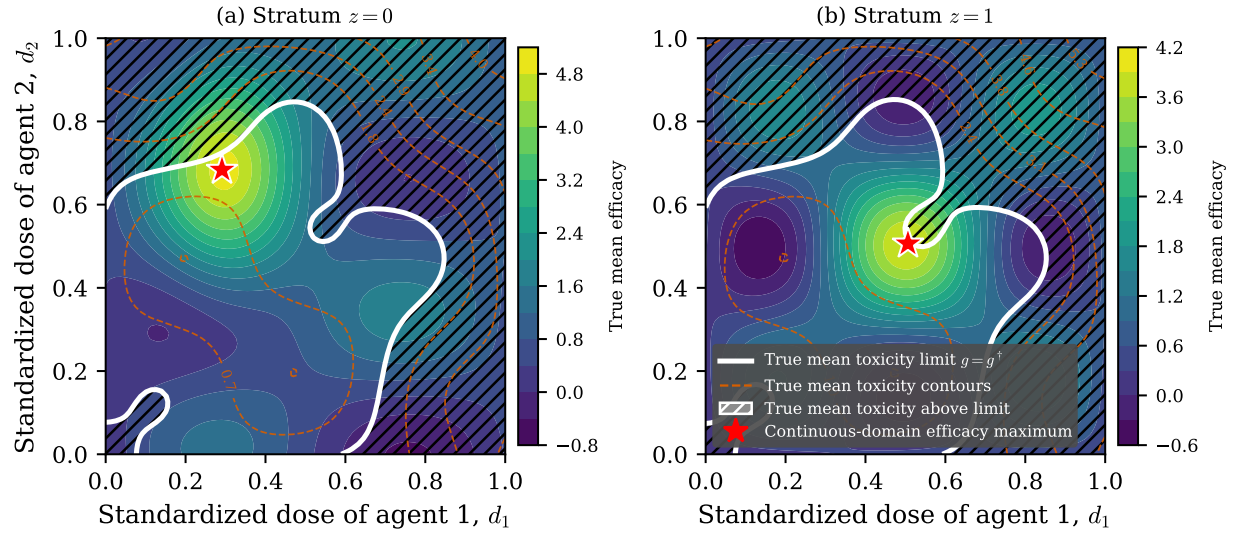


Figure 5: Gaussian-bump true mean efficacy and toxicity by stratum. Filled contours show efficacy, dashed orange contours show toxicity, the white contour marks $g = g^\dagger$, and hatching marks combinations above the limit. The red star is the fine-grid approximation to the continuous-domain unconstrained efficacy maximum. Doses are standardized to $[0, 1]$; color scales are panel-specific, and the 5×5 candidate grid is omitted.